

DYNAMICAL SYSTEMS

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Exam 2 - Solutions

Question Q1. (10 points) **Linear Stability and Bifurcation.** Consider the two-dimensional dynamical system depending on a real parameter $p \in \mathbb{R}$:

$$\begin{aligned}x' &= (p - y^2)x - y, \\y' &= x + (p - x^2)y.\end{aligned}$$

1. (3 points) Show there exist $\varepsilon > 0$ and $\delta > 0$ such that for every $|p| < \varepsilon$ the point $(0, 0)$ is the unique equilibrium point of the above system lying within $\{(x, y) \in \mathbb{R}^2 : \|(x, y)\| \leq \delta\}$.
Hint: Use the implicit function theorem.
2. (4 points) For each $p \in \mathbb{R}$, compute the linear stability of the point $(0, 0)$ and determine the value(s) of p at which a bifurcation occurs and identify the type of bifurcation.
3. (3 points) Sketch the phase portrait of the system for $p < 0$, $p = 0$, and $p > 0$, highlighting the equilibrium and any periodic orbits that may appear.

Solution:

1. Consider the right hand side as a function $F : \mathbb{R}^3 \rightarrow \mathbb{R}^2; (p, x, y) \mapsto (F_1(p, x, y), F_2(p, x, y))$ and consider its derivative along the spatial coordinates:

$$J_p(x, y) = \begin{pmatrix} \frac{\partial F_1}{\partial x} & \frac{\partial F_1}{\partial y} \\ \frac{\partial F_2}{\partial x} & \frac{\partial F_2}{\partial y} \end{pmatrix} = \begin{pmatrix} p - y^2 & -2xy - 1 \\ -2xy + 1 & p - x^2 \end{pmatrix}.$$

For $p = 0$ and at the equilibrium point $(x, y) = (0, 0)$ we have

$$DF(0, 0, 0) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \det(DF(0, 0, 0)) = 1 \neq 0$$

1 pt. for setup

Thus by the implicit function theorem there exists an open neighborhood $\mathcal{U} \subseteq \mathbb{R}$ containing 0, and an open neighborhood $\mathcal{V} \subseteq \mathbb{R}^2$ containing $(0, 0)$ and a unique function $(\tilde{x}, \tilde{y}) : \mathcal{U} \rightarrow \mathbb{R}^2$, which is differentiable, such that $(\tilde{x}(0), \tilde{y}(0)) = (0, 0)$ and for every $p \in \mathcal{U}$ the point $(p, \tilde{x}(p), \tilde{y}(p))$ satisfies

$$F(p, \tilde{x}(p), \tilde{y}(p)) = 0;$$

1 pt. for application

i.e. the unique equilibrium point of the system. But since for every $p \in \mathbb{R}$ the point $(0, 0)$ is an equilibrium point, by uniqueness, we have $(\tilde{p}(0), \tilde{p}(0)) = (0, 0)$ for every $p \in \mathcal{U}$. Choosing $\delta > 0$ small enough such that the δ -ball is contained in $\mathcal{V}$ and $\varepsilon > 0$ small enough (via continuity), the claim follows.

1 pt. for conclusion

2. The Jacobian matrix is

$$J_p(x, y) = \begin{pmatrix} \frac{\partial F_1}{\partial x} & \frac{\partial F_1}{\partial y} \\ \frac{\partial F_2}{\partial x} & \frac{\partial F_2}{\partial y} \end{pmatrix} = \begin{pmatrix} p - y^2 & -2xy - 1 \\ -2xy + 1 & p - x^2 \end{pmatrix}.$$

Evaluating at the origin $(0, 0)$ gives

$$J(0, 0) = \begin{pmatrix} p & -1 \\ 1 & p \end{pmatrix}.$$

1 pt. for Jacobian

For the Eigenvalues we solve $\det(J - \lambda I) = 0$:

$$\det \begin{pmatrix} p - \lambda & -1 \\ 1 & p - \lambda \end{pmatrix} = (p - \lambda)^2 + 1 = 0 \implies \lambda = p \pm i.$$

Classification:

- $p < 0$: $\operatorname{Re}(\lambda) < 0 \rightarrow$ stable spiral.
- $p = 0$: purely imaginary eigenvalues $\pm i \rightarrow$ center.
- $p > 0$: $\operatorname{Re}(\lambda) > 0 \rightarrow$ unstable spiral.

1 pt. for linear stability

1 pt. for bifurcation point

According to Theorem 8.10 in the lecture notes, this is a Hopf bifurcation. Since the transition is from stable to unstable, it is of subcritical type.

1 pt. for bifurcation type

3. The phase portrait can be sketched from the knowledge of linear classification above and the existence of a limit cycle after Hopf bifurcation.

1 pt. for identifying rotation

1 pt. for correct attraction

1 pt. for rest

Question Q2. (10 points) **Center manifold.** Consider the system

$$\begin{aligned} x' &= x^2 - y^2 \\ y' &= y - x^2 \end{aligned} \tag{1}$$

where $(x, y) \in \mathbb{R}^2$.

1. (2 points) Draw the nullclines and use them to sketch the flow in the vicinity of the y -nullcline (approximately for $-1 < x < 1$, $-1 < y < 1$)

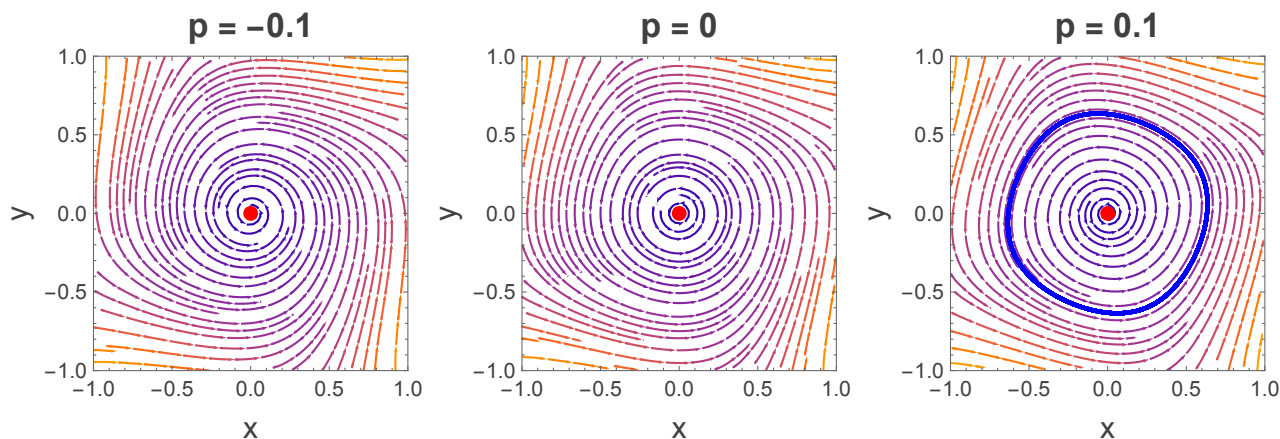


Figure 1: Sketch of bifurcation

2. (2 points) Compute the equilibrium points and state their linearized stability type.
3. (4 points) Compute an approximation of the center manifold of the origin via a degree 3 ansatz and conclude whether or not the origin is (non-linearly) stable.
4. (2 points) Does the system induce a Morse–Smale flow?
5. (2 points) State Peixoto’s theorem.

Solution:

1. **x-nullcline:** $x' = 0 \implies x^2 - y^2 = 0 \implies y = \pm x$

y-nullcline: $y' = 0 \implies y - x^2 = 0 \implies y = x^2$

Flow near y-nullcline $y = x^2$ for $-2 < x < 2$:

- Above $y = x^2$: $y' = y - x^2 > 0 \implies y$ increases.
- Below $y = x^2$: $y' < 0 \implies y$ decreases.

From drawing the nullclines we see that the y -nullcline lies in the region $|y| < |x|$ and in that region $x' = x^2 - y^2 > 0$ implies x increases.

1 pt. sketching nullclines

1 pt. sketching flow

2. Solve $x' = 0$ and $y' = 0$: Plugging the second equation into the first gives

$$x^2 - (x^2)^2 = x^2(1 - x^2) = 0$$

Thus $x = 0, \pm 1$ and, via $y = x^2$, the solutions are

$$(0, 0), \quad (1, 1), \quad (-1, 1)$$

1 pt. eq. points Jacobian matrix:

$$J(x, y) = \begin{pmatrix} 2x & -2y \\ -2x & 1 \end{pmatrix}$$

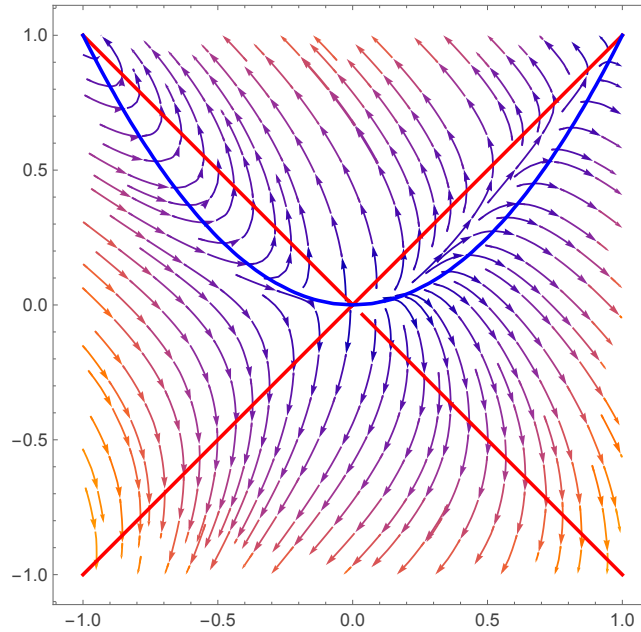


Figure 2: x -nullcline in red, y -nullcline in blue

At (0,0):

$$J(0,0) = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad \det = 0, \text{Tr} = 1, \lambda = 0, 1 \implies \text{non-hyperbolic/unstable}$$

At (1,1):

$$J(1,1) = \begin{pmatrix} 2 & -2 \\ -2 & 1 \end{pmatrix}, \quad \det = -2, \text{Tr} = 3, \lambda = \frac{3 \pm \sqrt{17}}{2} \implies \text{saddle (unstable)}$$

At (-1,1):

$$J(-1,1) = \begin{pmatrix} -2 & -2 \\ 2 & 1 \end{pmatrix}, \quad \det = 2, \text{Tr} = -1, \lambda = \frac{-1 \pm i\sqrt{7}}{2} \implies \text{stable focus}$$

1 pt. linear stability

3. Let's carry out a 3rd order approximation of the center manifold for the origin (note that the system is in normal form with $B = 0$ and $C = 1$). Ansatz: $y = h(x) = a_2x^2 + a_3x^3 + O(x^4)$. Then, for the invariance formula for the center manifold, we obtain **1 pt. for Ansatz**

$$\begin{aligned} 0 &= (2a_2x + 3a_3x^2 + O(x^3))(x^2 + O(x^4)) - a_2x^2 - a_3x^3 + x^2 \\ 0 &= 2a_2x^3 + O(x^4) - a_2x^2 - a_3x^3 + x^2 \\ 0 &= x^2(1 - a_2) + x^3(2a_2 - a_3) + O(x^4) \end{aligned}$$

1 pt. for invariance formula Therefore, we impose $a_2 = 1$ and $a_3 = 2$ and obtain the approximation $y \sim h(x) = x^2 + 2x^3$. **1 pt. for approximation** Then, thanks to the center manifold reduction theorem we have that (2) is locally topologically equivalent to the system

$$\begin{aligned} x' &= Bx + F(x, h(x)) = x^2 - x^4 - 4x^5 - 4x^6 + O(x^7) \\ y' &= y \end{aligned}$$

in a neighborhood of the origin. Therefore, the origin is nonlinearly unstable. **1 pt. for conclusion of stability**

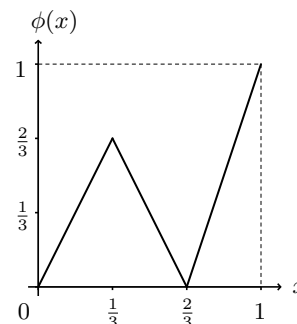
4. No, because the origin is not a hyperbolic equilibrium - a violation of Definition 6.7 in the lecture notes. **2 pt. for answer + justification**
5. See Theorem 6.6 in the lecture notes. **2 pt. for statement**

Question Q3. (4 points) **Chaos.**

Let $\phi : [0, 1] \rightarrow [0, 1]$ be a map and let $([0, 1], (\phi^n)_{n \in \mathbb{N}}, \mathbb{N})$ be the discrete time dynamical system induced by ϕ .

1. (2 points) State the definition of topological transitivity for this dynamical system.
2. (2 points) State the definition of sensitive initial condition for this dynamical system.
3. (4 points) Consider the map $\phi : [0, 1] \rightarrow [0, 1]$, given by

$$\phi(x) = \begin{cases} 2x & \text{if } x \in [0, \frac{1}{3}], \\ \frac{4}{3} - 2x & \text{if } x \in [\frac{1}{3}, \frac{2}{3}], \\ 3x - 2 & \text{if } x \in [\frac{2}{3}, 1]. \end{cases}$$



Show that ϕ is not chaotic.

Solution:

1. A dynamical system as specified above is topologically transitive if there exists $x \in [0, 1]$ such that its forward orbit $\mathcal{O}^+(x) := \{\phi^n(x) : n \in \mathbb{N}\}$ is dense in $[0, 1]$. **1 pt. for mostly correct, 2 pts. for fully correct**
2. A dynamical systems as specified above has sensitive dependence on initial conditions if for any $x_0 \in [0, 1]$ there exists $r > 0$ such that for any $\delta > 0$, there exists $y_0 \in [0, 1]$ with $|x_0 - y_0| < \delta$ and $n \in \mathbb{N}$ such that

$$|\phi^n(x_0) - \phi^n(y_0)| \geq r.$$

1 pt. for mostly correct, 2 pts. for fully correct

3. We show that the map ϕ is not topologically transitive, i.e. no point has dense forward orbit. First observe that $\phi([0, \frac{2}{3}]) = [0, \frac{2}{3}]$, so the interval $[0, \frac{2}{3}]$ is invariant. Now let $x \in [0, 1]$ be arbitrary and define the forward orbit $\mathcal{O}^+(x) = \{\phi^n(x) : n \in \mathbb{N}\}$. There are two possibilities:
 1. Either $\mathcal{O}^+(x) \in (\frac{2}{3}, 1]$. Then the closure of $\mathcal{O}^+(x) \subseteq [\frac{2}{3}, 1]$ and hence $\mathcal{O}^+(x)$ is not dense.
 2. There is a $k \in \mathbb{N}$ such that $\phi^k(x) \in [0, \frac{2}{3}]$. Since $[0, \frac{2}{3}]$ is invariant, all subsequent elements of the orbit also lie in the interval $[0, \frac{2}{3}]$. This means, that there are at most k points of the orbits lying in $(\frac{2}{3}, 1]$, which cannot be dense in $(\frac{2}{3}, 1]$.

This shows that no orbit can be dense, i.e. ϕ is not topologically transitive and thus not chaotic.

1 pt. for which property is violated

1 pt. for argument in $[0, 2/3]$

1 pt. for argument in $(2/3, 1]$

1 pt. for conclusion

generally 3 out of 4 pts. if argument sketchy, but mostly correct

Question Q4. (10 points) **Mixing.**

- a) (5 points) Prove that maps g_n , with $n \in \mathbb{N}$, from the unit circle $\mathbb{S}^1 := \{z \in \mathbb{C} \mid |z| = 1\}$ into itself of the form $z \mapsto g_n(z) = z^n$ preserve the Lebesgue measure/uniform distribution. *Hint: You can assume without proof that it is sufficient to show the claim on intervals.*
- b) (5 points) Prove that the maps g_n , with $n \in \mathbb{N}$, are strongly mixing.

Solution:

- a) Fix $n \in \mathbb{N}$ and denote $g_n : \mathbb{S}^1 \rightarrow \mathbb{S}^1$, $z \mapsto g_n(z) = z^n$. We shall read $\mathbb{S}^1$ as $[0, 2\pi) \bmod 2\pi$, and for any set $\{e^{i\theta} \mid \theta \in A \subset [0, 2\pi)\}$, define the induced Lebesgue measure on $\mathbb{S}^1$ as

$$\mu_{\mathbb{S}^1}(\{e^{i\theta} \mid \theta \in A \subset [0, 2\pi)\}) = \mu(A),$$

where μ is the Lebesgue measure on $\mathbb{R}$. Since the set of intervals $[a, b) \subset [0, 2\pi)$ is an intersection stable generator for the Borel σ -algebra on $[0, 2\pi)$, by the hint, it suffices to carry out the analysis only on such intervals. Note that

$$\mu_{\mathbb{S}^1}(\{e^{i\theta} \mid \theta \in [\theta_0, \theta_1) \subset [0, 2\pi)\}) = \mu([\theta_0, \theta_1)) = \theta_1 - \theta_0.$$

Now,

$$\begin{aligned} g_n^{-1}(\{e^{i\theta} \mid \theta \in [\theta_0, \theta_1) \subset [0, 2\pi)\}) &= \left\{ \sqrt[n]{e^{i\theta}} \mid \theta \in [\theta_0, \theta_1) \subset [0, 2\pi) \right\} \\ &= \left\{ e^{i\theta} \mid \theta \in \left[\frac{\theta_0 + 2k\pi}{n}, \frac{\theta_1 + 2k\pi}{n} \right), k = 0, \dots, n-1 \right\} \end{aligned}$$

and consequently

$$\begin{aligned} \mu_{\mathbb{S}^1}(g_n^{-1}(\{e^{i\theta} \mid \theta \in [\theta_0, \theta_1) \subset [0, 2\pi)\})) &= \sum_{k=0}^{n-1} \mu\left(\left[\frac{\theta_0 + 2k\pi}{n}, \frac{\theta_1 + 2k\pi}{n}\right)\right) = \theta_1 - \theta_0 \\ &= \mu_{\mathbb{S}^1}(\{e^{i\theta} \mid \theta \in [\theta_0, \theta_1) \subset [0, 2\pi)\}). \end{aligned}$$

1 pt. setup for measure

2 pt. showing special case/demonstrating understanding of the dynamics

1 pt. showing full case

1 pt. conclusion

- b) We shall prove that the maps g_n , $n \in \mathbb{N}$, are strong-mixing. Consider $h : [0, 1]/(0 \sim 1) \rightarrow \mathbb{S}^1 \subset \mathbb{C}$, $x \mapsto e^{i2\pi x}$. Note that h is a homeomorphism and that $h^{-1} \circ g_n \circ h : [0, 1]/(0 \sim 1) \rightarrow [0, 1]/(0 \sim 1)$, $x \mapsto h^{-1} \circ g_n \circ h(x) = nx \bmod 1$, that is, $h^{-1} \circ g_n \circ h$ is an expanding circle endomorphism and therefore strong mixing by Proposition 22.5 in the lecture notes.

For any $\mathcal{A} \in \mathcal{F}$ Borel σ -algebra on $\mathbb{S}^1 \subset \mathbb{C}$ and taken A the Borel set $A = h^{-1}(\mathcal{A})$, using the definition of $\mu_{\mathbb{S}^1}$ in a), we have that $\mu(A) = \mu_{\mathbb{S}^1}(\mathcal{A})$. Now let, $\mathcal{A}, \mathcal{B} \in \mathcal{F}$ Borel σ -algebra on $\mathbb{S}^1 \subset \mathbb{C}$ and A, B the Borel sets $A = h^{-1}(\mathcal{A}), B = h^{-1}(\mathcal{B})$. Gathering all together, we obtain that

$$\begin{aligned}
\lim_{k \rightarrow \infty} \mu_{\mathbb{S}^1}(g_n^k(\mathcal{A}) \cap \mathcal{B}) &= \lim_{k \rightarrow \infty} 2\pi \mu(h^{-1}(g_n^k(h(A)) \cap (h(B))) \\
&= \lim_{k \rightarrow \infty} 2\pi \mu((h^{-1} \circ g_n \circ h)^k(A) \cap B) = \mu(A)\mu(B) = \mu_{\mathbb{S}^1}(\mathcal{A})\mu_{\mathbb{S}^1}(\mathcal{B}),
\end{aligned}$$

that is, also g_n is strong-mixing.

1 pt. for setup

2 pt. for main argument

1 pt. connecting to Prop. 22.5 or similar

1 pt. for conclusion